

Lab 8

ANOVA: Between-Groups and Within-Groups

Walkthrough (5 points)

Instructions

We'll be using the `superman_smes_data.sav` dataset to practice with ANOVA analyses.

Researchers surveyed 47 people who watched Superman films and measured their movie enjoyment using the Subjective Movie Enjoyment Scale (SMES).

- **Height gap (2 categories)** (`height_gap_2cat`) — Values: 1 = Small, 2 = Big
- **Emotional Impact subscale score** (`emotional_impact`)

RH1: Between-Groups ANOVA Walkthrough

Test the research hypothesis that films with a big height gap between actors (`height_gap` = 3) have higher emotional impact scores (`emotional_impact`) than films with a small height gap (minimal or average, `height_gap` = 1 or 2).

The `superman_smes_data.sav` dataset includes the following variables you will use:

- **Height gap (2 categories)** (`height_gap_2cat`) — Values: 1 = Small, 2 = Big
- **Emotional Impact subscale score** (`emotional_impact`)

Data Analysis:

1. **Considering the RH:** Does this research hypothesis suggest that one group will have higher scores, lower scores, or equal scores compared to the other group? Which group is expected to have higher scores?

Answer:

2. **Open the dataset (`superman_smes_data.sav`):** Make sure that...

- Each variable has a label
- All qualitative variables have value labels
- All variables have missing values declared as `-99`

3. **Data Transformation:** The hypothesis wants to compare films with big height gaps versus films with small height gaps (minimal or average combined). We need to recode `height_gap` into a new variable with only two categories.

- Recode `height_gap` into a new variable called `height_gap_2cat`
 - Small height gap = 1 (includes original values 1 = Minimal and 2 = Average)
 - Big height gap = 2 (includes original value 3 = Big)

4. **Data Selection:** No selection needed for this analysis.

5. **Data Analysis:** Follow the handout to compute an Independent Groups ANOVA with `height_gap_2cat` as the IV and `emotional_impact` as the DV.

RH1 Results**Descriptive Statistics:**

height_gap_2cat	<i>n</i>	<i>M</i>	<i>SD</i>
Small Height Gap	_____	_____	_____
Big Height Gap	_____	_____	_____

ANOVA:

<i>F</i>	<i>df</i>	<i>MSE</i>	<i>p</i>
_____	_____, _____	_____	_____

6. State the H_0 :

H_0 :

7. Decision:

Retain or reject H_0 ?

Support research hypothesis?

8. Write-up with univariate stats:

Write-up:

9. Grab a screenshot of the homework checker screen, showing all of your input answers in green (correct). Paste it below this step. **Your TA will not grade the assignment without this.**

Paste Homework Checker screenshot here:

RH2: Within-Groups ANOVA Walkthrough

Test the research hypothesis that emotional impact scores (*emotional_impact*) are higher than aesthetic appeal scores (*aesthetic_appeal*) for Superman films.

The *superman_smes_data.sav* dataset includes the following variables you will use:

- **Emotional Impact** (*emotional_impact*) – continuous scale score
- **Aesthetic Appeal** (*aesthetic_appeal*) – continuous scale score

Data Analysis:

1. **Considering the RH:** Does this research hypothesis suggest that one condition will have higher scores, lower scores, or equal scores compared to the other condition? Which condition is expected to have higher scores?

Answer:

2. **Open the dataset (superman_smes_data.sav):** Make sure that...

- Each variable has a label
- Missing values are declared for both SMES subscale variables

3. **Data Transformation:** No transformation is needed for these data - both DV scores (emotional_impact and aesthetic_appeal) are already computed.

4. **Data Selection:** No selection needed for this analysis.

5. **Data Analysis:** Follow the handout to compute a Dependent Groups ANOVA with emotional_impact and aesthetic_appeal as the variables in the analysis.

RH2 Results**Descriptive Statistics:**

Condition	<i>M</i>	<i>SD</i>	<i>n</i>
Emotional Impact	_____	_____	_____
Aesthetic Appeal	_____	_____	_____

Repeated Measures ANOVA:

<i>F</i>	<i>df</i>	<i>MSE</i>	<i>p</i>
_____	_____, _____	_____	_____

6. **State the H_0 :**

H_0 :

7. **Decision:**

Retain or reject H_0 ?

Support research hypothesis?

8. **Write-up with univariate stats:**

Write-up:

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Multigroup Analysis (15 points)

Instructions

Another important skill is to be able to identify the proper variables and select and complete the proper analysis for questions about a dataset with several variables. In this section, we'll use the `hotones_data.sav` dataset to practice identifying which type of ANOVA to use for different research questions.

A researcher has collected data about Hot Ones episodes to test various hypotheses about the show. The dataset includes information about YouTube engagement metrics, guest characteristics, and the Scoville Heat Units (SHU) of each hot sauce used in the episodes.

Open the dataset and identify which variables are qualitative and which are quantitative.

A researcher has collected these data to test various hypotheses about Hot Ones episodes. Below are four questions the researcher has asked you to test. For each, identify the appropriate type of ANOVA, the IV and the DV, write the H_0 , and tell which variables you will use to complete the analysis.

RH3: Multigroup Analysis Question #1

Research Question: Among episodes where guests completed the challenge without quitting, do episodes with the first hot sauce (`SHU_1`) have lower Scoville Heat Units than the tenth hot sauce (`SHU_10`)?

The `hotones_data.sav` dataset includes the following variables you will use:

- **First Sauce SHU in millions (`SHU_1_mil`)** – continuous
- **Tenth Sauce SHU in millions (`SHU_10_mil`)** – continuous

1. Type of ANOVA:

Between-Groups (BG) or Within-Groups (WG)?

2. Independent Variable (IV):

Answer:

3. Dependent Variable (DV):

Answer:

4. State the H_0 :

H_0 :

5. Data Transformation and Selection: The SHU values are very large numbers. To make them easier to work with, we need to convert them to millions of SHU. Additionally, the research question specifies “episodes where guests completed the challenge without quitting.”

- **Select Cases:** Select only episodes where `result = 1` (completed challenge)
- Create a new variable $SHU_1_mil = SHU_1 / 1000000$
- Create a new variable $SHU_10_mil = SHU_10 / 1000000$

6. Variables you will use:

Answer:

7. Data Analysis: Follow the handout to compute a Dependent Groups ANOVA with `SHU_1_mil` and `SHU_10_mil` as the variables in the analysis.

RH3 Results

Descriptive Statistics:

Condition	<i>M</i>	<i>SD</i>	<i>n</i>
First Sauce (millions SHU)	_____	_____	_____
Tenth Sauce (millions SHU)	_____	_____	_____

Repeated Measures ANOVA:

<i>F</i>	<i>df</i>	<i>MSE</i>	<i>p</i>
_____	_____, _____	_____	_____

8. Decision:

Retain or reject H_0 ?

Support research hypothesis?

9. Write-up with univariate stats:

Write-up:

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RH4: Multigroup Analysis Question #2

Research Question: Among episodes where guests completed the challenge without quitting, do actors receive more YouTube views (*views*) than non-actors? Note: the *views* variable is already recorded in millions.

The *hotones_data.sav* dataset includes the following variables you will use:

- **Guest occupation type (*actor*)** - categorical/nominal; 2-levels
- **YouTube views in millions (*views*)** - continuous (already in millions in the dataset)

1. Type of ANOVA:

Between-Groups (BG) or Within-Groups (WG)?

2. Independent Variable (IV):

Answer:

3. Dependent Variable (DV):

Answer:

4. State the H_0 :

H_0 :

5. Data Transformation and Selection: We need to create an *actor* variable from the *occupation* variable. Additionally, the research question specifies “episodes where guests completed the challenge without quitting.”

- **Select Cases:** Select only episodes where *result* = 1 (completed challenge)
- Create a new variable *actor* where:
 - *actor* = 1 if *occupation* is 3, 4, or 7 (actors)
 - *actor* = 0 for all other occupations (non-actors)

6. Variables you will use:

Answer:

7. Data Analysis: Follow the handout to compute an Independent Groups ANOVA with *actor* as the IV and *views* as the DV.

RH4 Results

Descriptive Statistics:

actor	<i>n</i>	<i>M</i>	<i>SD</i>
Non-Actor	_____	_____	_____
Actor	_____	_____	_____

ANOVA:

<i>F</i>	<i>df</i>	<i>MSE</i>	<i>p</i>
_____	_____	_____	_____

8. Decision:

Retain or reject H_0 ?

Support research hypothesis?

9. Write-up with univariate stats:

Write-up:

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RH5: Multigroup Analysis Question #3

Research Question: Do episodes receive more comments than likes? Because the raw counts are very large numbers, first transform the variables into thousands by computing `comments_thous = comments / 1000` and `likes_thous = likes / 1000`, then run the analysis on the transformed variables.

The `hotones_data.sav` dataset includes the following variables you will use:

- **Comments in thousands (`comments_thous`)** – continuous (computed from `comments / 1000`)
- **Likes in thousands (`likes_thous`)** – continuous (computed from `likes / 1000`)

1. Type of ANOVA:

Between-Groups (BG) or Within-Groups (WG)?

2. Independent Variable (IV):

Answer:

3. Dependent Variable (DV):

Answer:

4. State the H_0 : H_0 :

5. Data Transformation: Because the raw counts are very large numbers, transform the variables into thousands: compute `comments_thous = comments / 1000` and `likes_thous = likes / 1000`.

6. Data Selection: No selection needed for this analysis - we are examining all episodes.

7. Variables you will use:

Answer:

8. Data Analysis: Follow the handout to compute a Dependent Groups ANOVA with `comments_thous` and `likes_thous` as the variables in the analysis.

RH5 Results**Descriptive Statistics:**

Condition	<i>M</i>	<i>SD</i>	<i>n</i>
Comments (thousands)	_____	_____	_____
Likes (thousands)	_____	_____	_____

Repeated Measures ANOVA:

<i>F</i>	<i>df</i>	<i>MSE</i>	<i>p</i>
_____	_____, _____	_____	_____

9. Decision:Retain or reject H_0 ?

Support research hypothesis?

10. Write-up with univariate stats:

Write-up:

11. Grab a screenshot of the homework checker screen, showing all of your input answers in green (correct). Paste it below this step. **Your TA will not grade the assignment without this.**

Paste Homework Checker screenshot here:

RH6: Multigroup Analysis Question #4

Research Question: Among episodes where guests completed the challenge without quitting, do younger guests (under 35) receive more YouTube likes than older guests (35 and over)? Because the raw like counts are very large numbers, first transform the variable into thousands by computing $\text{likes_thous} = \text{likes} / 1000$, then run the analysis on the transformed variable.

The hotones_data.sav dataset includes the following variables you will use:

- **Guest age group (age_grp)** - categorical/nominal; 2-levels
- **YouTube likes in thousands (likes_thous)** - continuous (computed from $\text{likes} / 1000$)

1. Type of ANOVA:

Between-Groups (BG) or Within-Groups (WG)?

2. Independent Variable (IV):

Answer:

3. Dependent Variable (DV):

Answer:

4. State the H_0 :

H_0 :

5. Data Transformation and Selection: We need to create an age_grp variable from the age variable, and transform likes into thousands. Additionally, the research question specifies “episodes where guests completed the challenge without quitting.”

- **Select Cases:** Select only episodes where $\text{result} = 1$ (completed challenge)
- Transform likes into thousands: compute $\text{likes_thous} = \text{likes} / 1000$
- Create a new variable age_grp where:
 - age_grp = 1 if age < 35 (younger guests)
 - age_grp = 2 if age ≥ 35 (older guests)

6. Variables you will use:

Answer:

7. Data Analysis: Follow the handout to compute an Independent Groups ANOVA with `age_grp` as the IV and `likes_thous` as the DV.

RH6 Results

Descriptive Statistics:

age_grp	<i>n</i>	<i>M</i>	<i>SD</i>
Under 35	_____	_____	_____
35 and Over	_____	_____	_____

ANOVA:

<i>F</i>	<i>df</i>	<i>MSE</i>	<i>p</i>
_____	_____	_____	_____

8. Decision:

Retain or reject H_0 ?

Support research hypothesis?

9. Write-up with univariate stats:

Write-up:

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